Common Mistakes

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Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# printr is not loaded here: it inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
```

```
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
# Summarizes key information about statistical objects in tidy tibbles
library(broom)
# Provides suite of functions to work with regression model
# `broom::tidy()` tibbles
library(broom.helpers)
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")
    )
)

knitr::opts_chunk$set(message = FALSE)
options(digits = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
# use the `filter()` function from dplyr() by default
conflicts_prefer(dplyr::filter)
legend_text_size <- 9
run_graphs <- TRUE
```

1 Parameters versus random variables

The parameters of a probability distribution shouldn't involve the random variables[1] being modeled:

⚠ Incorrect parameter dependency on random variables

$$X \sim Pois(\lambda)$$
$$\hat{\lambda}_{ML} \to_D N(\bar{X}, \lambda/n)$$

Solution

Solution.

$$\hat{\lambda}_{ML} \to_D N(\lambda, \lambda/n)$$

Expectations[2] are means, not sums, despite the similarity of Σ and E. Really, we should use μ instead of E.

[1] https://morrison-lab.github.io/pds/random-variables.html#def-random-variable
[2] https://morrison-lab.github.io/pds/expectation.html#def-expectation

2 Regression modeling

Each mistake in this section is shown in a warning callout, followed by a corrected version.

2.1 Logistic-regression coefficients are log odds ratios

In a logistic regression model, a coefficient β_j is a difference in log-odds[3], so $\exp\{\beta_j\}$ is an odds ratio[4] (see the odds ratio theorem for logistic regression[5]), not a risk ratio[6]:

⚠ Interpreting $\exp\{\beta_1\}$ as a risk ratio

"$\exp\{\beta_1\} = 1.5$, so a one-unit increase in x_1 makes the outcome 50% more likely."

Solution

Solution. "$\exp\{\beta_1\} = 1.5$, so a one-unit increase in x_1 multiplies the **odds** of the outcome by 1.5, holding the other covariates fixed: $\omega(\underset{\sim}{x}, \underset{\sim}{x}^*) = 1.5$ when $\underset{\sim}{x}$ and $\underset{\sim}{x}^*$ differ by one unit in x_1 only."

2.2 Odds ratios approximate risk ratios only for rare outcomes

The odds ratio[7] equals the risk ratio[8] times the correction factor $\frac{1-\pi_2}{1-\pi_1}$ (see the odds ratio correction-factor theorem[9]). That correction factor is close to 1 only when both risks are small.

⚠ Reporting an odds ratio as a risk ratio for a common outcome

$\pi_1 = 0.4$, $\pi_2 = 0.2$, odds ratio $= 2.67$: "exposed participants had 2.67 times the risk of unexposed participants."

Solution

Solution. The risk ratio is $\frac{\pi_1}{\pi_2} = \frac{0.4}{0.2} = 2$. The odds ratio is $\frac{0.4/0.6}{0.2/0.8} = \frac{0.667}{0.25} = 2.67$, which equals $2 \cdot \frac{0.8}{0.6}$. Report "2.67 times the odds", or estimate the risk ratio directly.

2.3 Poisson rate models need an offset

When subjects have different amounts of follow-up time t, a Poisson model for an event rate includes $\log\{t\}$ as an offset[10] (see the Poisson rate model's mean[11]). Without the offset term, the model describes expected counts, not rates, and $\exp\{\beta_j\}$ is a ratio of expected counts rather than a rate ratio[12].

⚠ Omitting the offset

Group A: 10 events in 1000 person-years; group B: 10 events in 500 person-years. A Poisson model fit to the counts alone gives $\exp\{\widehat{\beta}_1\} = 10/10 = 1$: "the groups have the same event rate."

[3] binary-outcome-associations.qmd#def-logodds
[4] binary-outcome-associations.qmd#def-OR
[5] logistic-regression.qmd#thm-logistic-OR
[6] binary-outcome-associations.qmd#def-RR
[7] binary-outcome-associations.qmd#def-OR
[8] binary-outcome-associations.qmd#def-RR
[9] binary-outcome-associations.qmd#thm-rare-disease-or-rr
[10] https://morrison-lab.github.io/pds/distributions.html#def-offset
[11] count-regression.qmd#eq-mean-poisson
[12] count-regression.qmd#thm-poisson-rate-ratio

Solution

Solution. Include $\log\{t\}$ as an offset (in R, `offset(log(t))` in the model formula). The incidence rates[a] are $10/1000 = 0.01$ and $10/500 = 0.02$ events per person-year, so the rate ratio for A versus B is $0.01/0.02 = 0.5$.

[a] intro-to-survival-analysis.qmd#def-incidence-rate

2.4 Hazard ratios are not ratios of survival probabilities

A hazard ratio[13] compares hazards[14], which are instantaneous event rates, not survival probabilities[15]. Under proportional hazards, $\mathrm{S}(t|\underset{\sim}{x}) = [\mathrm{S}_0(t)]^{\theta_\lambda(\underset{\sim}{x})}$ (see the proportional hazards survival theorem[16]), so the hazard ratio enters the survival function as an exponent, not as a multiplier.

⚠ Interpreting a hazard ratio as a survival-probability ratio

$\lambda(t|\underset{\sim}{x} : \underset{\sim}{x}^*) = 2$ and $\mathrm{S}(t|\underset{\sim}{x}^*) = 0.8$: "the exposed group has half the probability of surviving to t, $\mathrm{S}(t|\underset{\sim}{x}) = 0.4$."

Solution

Solution. $\mathrm{S}(t|\underset{\sim}{x}) = 0.8^2 = 0.64$, so the ratio of survival probabilities is $0.64/0.8 = 0.8$, and the ratio of cumulative risks is $\frac{1-0.64}{1-0.8} = \frac{0.36}{0.2} = 1.8$. Neither ratio equals the hazard ratio of 2.

2.5 Censored observations are not event-free

A right-censored[17] observation tells us only that the event had not occurred by the end of that subject's follow-up; the subject might have had the event later.

⚠ Treating censored observations as event-free

Of 100 participants, 20 had the event and 30 were censored before 5 years: "the 5-year risk is $20/100 = 0.2$, since the censored participants did not have the event."

Solution

Solution. Counting censored participants as event-free for the whole period biases the estimated risk downward. Use a survival-analysis method that keeps each censored participant in the risk set[a] only until their censoring time, such as the Kaplan-Meier estimator[b] or a proportional hazards model, under the assumption of non-informative censoring[c].

[a] intro-to-survival-analysis.qmd#def-risk-set
[b] intro-to-survival-analysis.qmd#def-KM-estimator
[c] intro-to-survival-analysis.qmd#def-noninformative-censoring

3 R

3.1 Don't copy-paste code

Successful programmers don't use copy-paste! Write functions instead.[18]

[13] intro-to-survival-analysis.qmd#def-hazard-ratio
[14] intro-to-survival-analysis.qmd#def-hazard
[15] intro-to-survival-analysis.qmd#def-surv-fn
[16] proportional-hazards-models.qmd#thm-ph-surv
[17] intro-to-survival-analysis.qmd#def-right-censor
[18] https://r4ds.hadley.nz/functions#introduction

4 Quarto

4.1 Separate divs and slide breaks

Make sure not to put a div `:::` on the next line after a slide break `---`:

```
---
::: notes
:::
```

There needs to be an empty line between them:

```
---

::: notes
:::
```

4.2 `library(printr)` currently breaks `df-print: paged`

See https://github.com/yihui/printr/issues/41

5 LaTeX

- don't use `align*` or `align*` in quarto; only `aligned`

Double superscript issues: https://www.overleaf.com/learn/latex/Errors/Double_superscript